Indoor Air Quality in Sport Facilities

Interdisciplinary Project Report

Course:	194.147 – Interdisciplinary Project in Data Science
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1 Introduction

Indoor air quality is often overlooked compared to outdoor pollution, but it is particularly important in enclosed sport environments like climbing halls, boulder gyms, and fitness centres. These spaces combine high occupancy, intense physical activity, and sometimes limited ventilation. These conditions can drive up CO_2, particulate matter (PM), and temperature to uncomfortable or harmful levels.

The goal of this project is to investigate how key air quality indicators, specifically CO_2, particulate matter (PM2.5), temperature, and humidity, vary between sport facility and home environments, and how they relate to occupancy and activity intensity. The dataset comes from a 17-day longitudinal study (June 29 to July 15, 2025). Twenty-two people answered daily well-being surveys, and 20 of them also have sensor data, from fixed and portable sensors at their home, at their workplace (a sport facility), or at both.

The analysis focuses on three questions: (1) How do CO_2, PM2.5, temperature and humidity differ between sport facilities and home? (2) Does activity intensity have a measurable effect on CO_2? (3) What temporal patterns emerge across the day and the study period?

2 Research Protocol

2.1 Participants and Study Design

Twenty-two participants answered the daily surveys. Sensor data exist for 20 of them (IDs between id088 and id108); the other two respondents have none. The sensors were set up in two places per participant, at home and at the workplace, which is a sport facility such as a gym or climbing hall. Ten of the 20 have fixed-sensor data from both places and the other ten from one place only. Data collection ran for about 17 days and produced 713 survey observations. Of these, 371 were matched to sensor data at work and 293 at home, and 49 could not be matched to a sensor location.

2.2 Sensor Setup, Calibration and Spatial Resolution

Two sensor types were deployed, providing complementary spatial coverage:

AirControl sensors (fixed placement): One sensor was placed at a fixed indoor position in each participant's home and one at their workplace. These provided continuous fixed-point monitoring of CO_2 (ppm), temperature (°C), and relative humidity (%) at high frequency, with timestamps logged in DD/MM/YYYY HH:mm:ss format. Data quality thresholds were applied as a calibration step: CO_2 readings outside 300–5000 ppm,

temperature outside 10–40°C, and humidity outside 10–90% were treated as implausible and removed.

AtmoTube sensors (mobile/portable): Participants carried these throughout the day. They captured a broader set of indicators: CO_2, temperature, humidity, VOC (volatile organic compounds, ppm), air quality score (AQS), and particulate matter PM1, PM2.5, and PM10 ($\mu g/m^3$). The portable nature of these sensors provided spatial coverage beyond fixed points, capturing air quality as participants moved through different areas of the facility or their home.

Together, the two sensor types provide both a stable fixed reference (AirControl) and a mobile picture of conditions the participant was actually exposed to (AtmoTube), giving a richer spatial and temporal resolution than either could provide alone.

2.3 Survey Data Collection

Participants completed daily surveys on a Likert scale (1–7) across five dimensions: stress, tiredness, productivity, thermal comfort (warmth), and mental demand. These subjective reports were timestamped, allowing them to be matched to the nearest sensor readings.

3 Methodology

3.1 Data Preprocessing

Raw data was loaded from 314 files across all participants. After applying the sensor thresholds described above, participant IDs were standardised and survey responses outside the valid 1–7 range were removed.

3.2 Timestamp Alignment

Sensor readings and survey responses were never recorded at exactly the same moment. To address this, **linear interpolation** was used to estimate the sensor value at the exact survey timestamp from the two nearest readings on either side:

$$v_t = v_{t_0} + \frac{t - t_0}{t_1 - t_0} \cdot (v_{t_1} - v_{t_0})$$

This is valid for physical measurements (CO_2, temperature, PM2.5) because they change continuously over time. Importantly, this was *not* applied to ordinal survey variables such as stress. A fractional value like "stress = 3.5" has no meaningful interpretation since the scale categories are not equally spaced.

3.3 Correlation Method

Spearman rank correlation was used throughout rather than Pearson. Survey variables (stress, tiredness, etc.) are ordinal and Pearson assumes equal-interval measurement, which cannot be assumed for Likert-scale data. Spearman converts values to ranks first, making it appropriate for mixed ordinal and continuous variable comparisons. As a check, Pearson coefficients were computed on the same 664 observations that have no missing values. They give the same picture: every correlation of about 0.3 or more in absolute value differs by at most 0.02 and keeps its sign. The largest difference in the whole matrix is 0.17, for CO_2 and humidity ($\rho \approx 0.04$ against $r \approx 0.21$), which are not part of the comparison with the survey. The values below are the Spearman coefficients computed by `correlation_analysis.py`.

4 Results

4.1 Air Quality Distributions

Figure 1 shows the distributions of the three AirControl sensor variables. CO_2 is right-skewed, with the bulk of readings falling below 600 ppm and a long tail reaching above 1400 ppm during peak periods. Temperature clusters around 26–28°C and humidity spans a wide range of 15–50%, reflecting variation across participants' homes and workplaces.

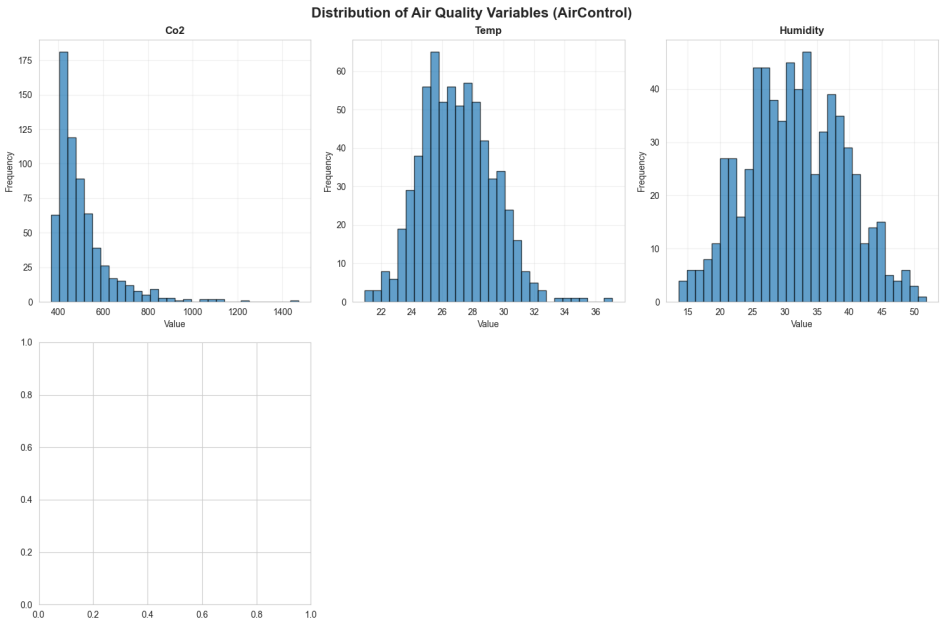


Figure 1: Distribution of AirControl sensor readings across all participants and both locations. CO_2 is right-skewed; temperature is approximately normal; humidity shows wide spread.

4.2 Work vs. Home: Air Quality and Particulate Matter

Table 1 summarises mean values for all key indicators by location type. Figure 2 shows the CO_2, temperature and humidity distributions visually.

Table 1: Mean air quality indicators by location type, evaluated at the survey timestamps (work $n = 371$, home $n = 293$ observations; PM2.5 from the portable sensors: work $n = 351$, home $n = 252$).

Indicator	**Work (Sport Facility)**	**Home**
CO_2 (ppm)	511.9	500.1
Temperature (°C)	27.0	26.9
Humidity (%)	30.9	32.2
PM2.5 ($\mu g/m^3$)	5.11	5.83

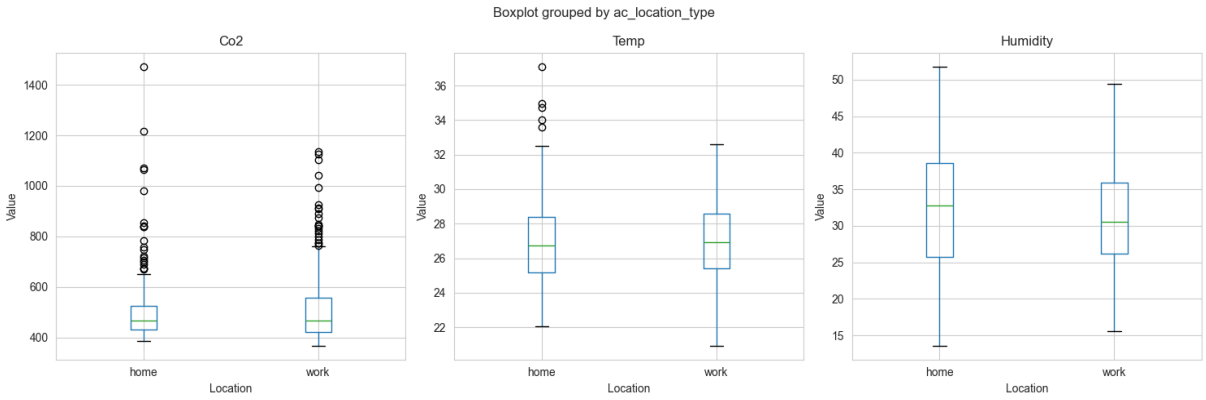


Figure 2: CO_2, temperature and humidity by location type. CO_2 is higher and more variable at work; temperature and humidity are nearly equal across both settings.

At the times participants completed their surveys, CO_2 was higher at work than at home: 511.9 against 500.1 ppm, a difference of about 12 ppm. Higher activity and occupancy in sport facilities could explain this, although occupancy was not measured. Readings above 800 ppm were slightly more common at work (17 of 371 observations, 4.6%) than at home (9 of 293, 3.1%), and the highest single reading, about 1470 ppm, came from a home. These means use survey timestamps only. The continuous AirControl record also covers the nights at home, and there the home mean is higher (566 against 448 ppm). The difference above therefore says nothing about total daily exposure. Temperature and humidity differ very little between the two settings.

Particulate matter tells a different story: PM2.5 is slightly *higher at home* (5.83 vs 5.11 $\mu g/m^3$). Both values are well below the WHO guideline of 15 $\mu g/m^3$, but the higher home

value is noteworthy. It may reflect cooking, outdoor air infiltration, or other household sources not present in the sport facility. This suggests that while sport facilities drive CO_2 up through occupancy and exertion, home environments contribute more to particulate exposure.

4.3 Survey Responses by Location

Figure 3 shows well-being scores at work vs. home. Stress is slightly higher at work (mean 3.48) compared to home (mean 3.25), but distributions overlap considerably. The other dimensions, including tiredness, productivity, warmth, and mental demand, are similar across both locations.

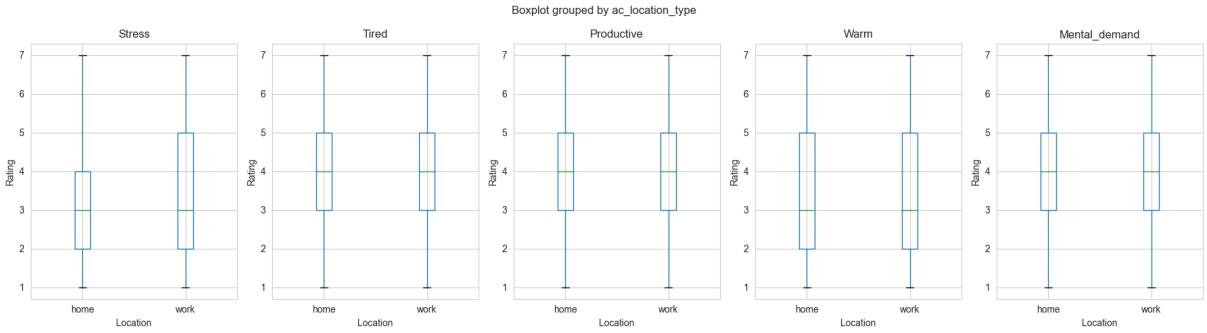


Figure 3: Survey response distributions by location. Stress is marginally higher at work; other dimensions are largely similar between settings.

4.4 Correlations: Air Quality and Well-being

Figure 4 shows the Spearman correlation matrix for the 664 observations without missing values. Between the measured air quality variables and the survey responses, the correlations are mostly weak (all $|\rho| \leq 0.18$). The exception is perceived warmth, which follows measured temperature ($\rho \approx 0.51$) and humidity ($\rho \approx -0.41$) and is also weakly related to CO_2 ($\rho \approx 0.21$). Among the survey variables, tiredness and productivity have the strongest relationship ($\rho \approx -0.39$), which is intuitive. Perceived warmth correlates moderately with tiredness ($\rho \approx 0.30$) and negatively with productivity ($\rho \approx -0.30$), while measured temperature itself is only weakly related to tiredness ($\rho \approx 0.07$). CO_2 does not correlate strongly with any single survey dimension ($|\rho| \leq 0.21$).

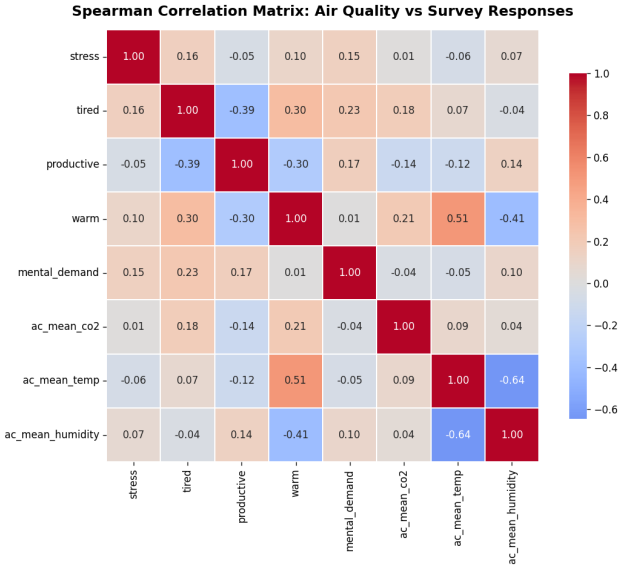


Figure 4: Spearman correlation matrix between AirControl air quality variables and survey responses ($n = 664$ complete observations). Most air quality and well-being correlations are weak; the strongest pattern is the tiredness–productivity relationship.

4.5 Temporal Patterns

Figure 5 shows daily patterns across the study period. CO_2 remains stable with no gradual buildup, suggesting consistent ventilation and stable conditions. Survey responses for stress and tiredness show more day-to-day variation, driven by individual factors rather than environmental drift.

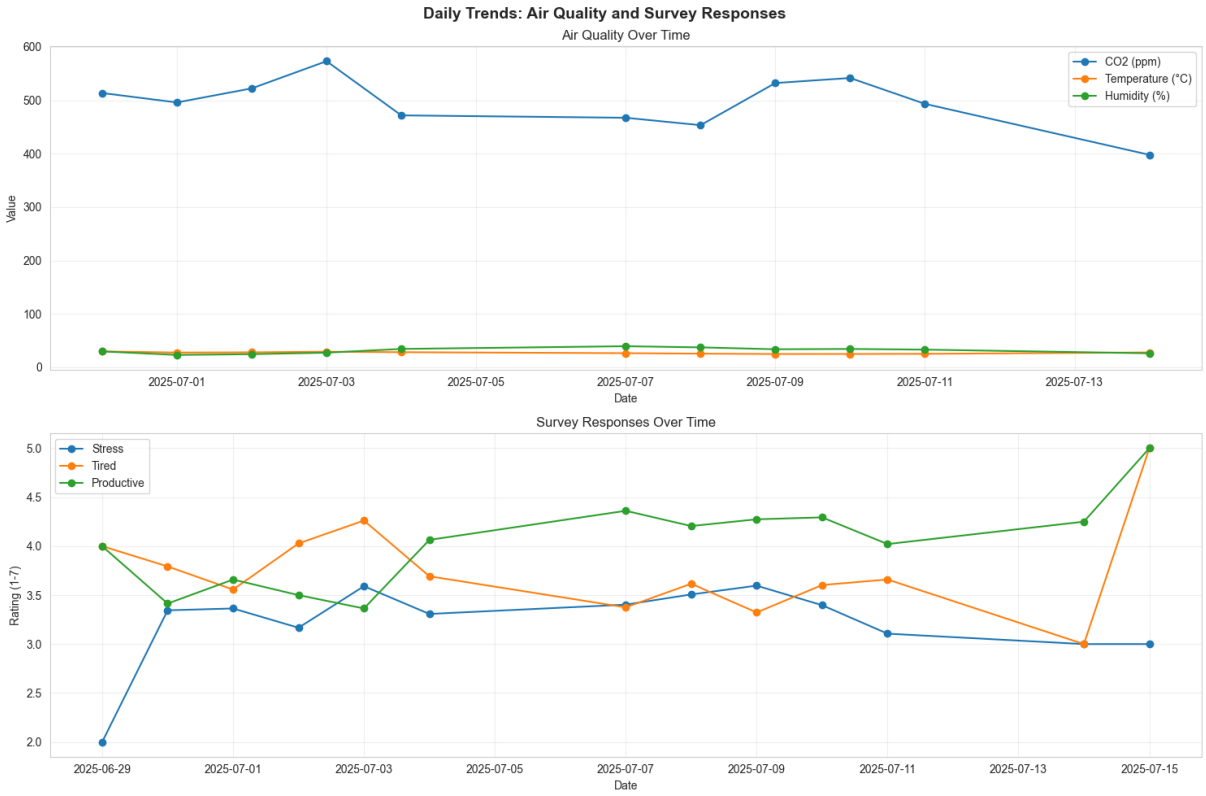


Figure 5: Daily trends across the 17-day study period. CO_2 is stable over time; survey responses show more variation driven by individual factors.

4.6 CO_2 Thresholds and Well-being

Grouping observations by CO_2 category (Low <800, Medium 800–1000, High >1000 ppm) reveals a hint that mental demand is slightly higher at elevated CO_2 levels (Figure 6). However, the Medium and High groups have few observations, so this should be interpreted cautiously.

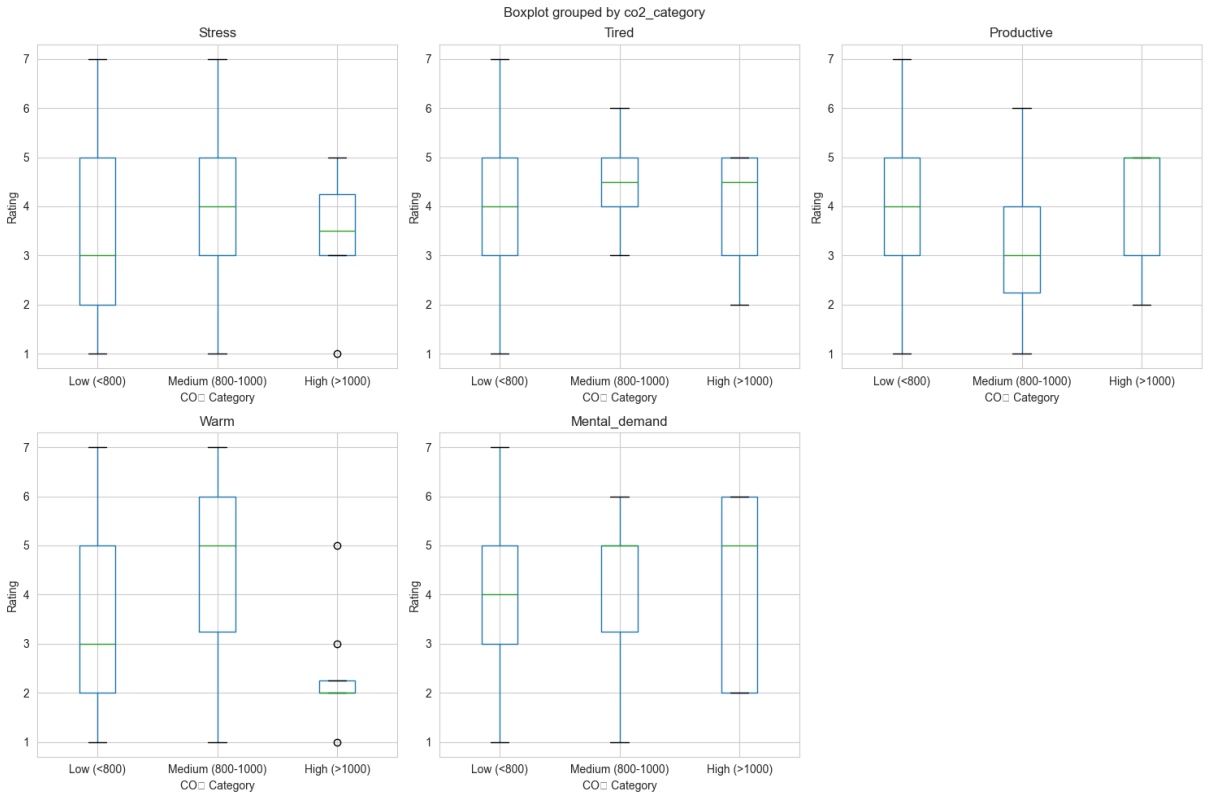


Figure 6: Survey responses by CO_2 category. Most observations fall in the Low category. Mental demand appears slightly elevated in the High CO_2 group, but sample sizes limit conclusions.

4.7 Activity Intensity and CO_2

The second research question asks whether activity intensity has a measurable effect on CO_2. Activity was not measured directly in this study, so a proxy was built from the survey answers: tiredness minus productivity, plus 7 so that the score stays positive. The score runs from 1 to 12 and is higher when a participant reports feeling more tired and less productive, as one would after more physical effort. It is a rough stand-in, and the results below depend on that assumption.

Across the 664 observations, the proxy is weakly and positively related to CO_2 (Spearman $\rho \approx 0.20$, Pearson $r \approx 0.16$, both $p < 0.001$). A simple linear regression gives about 8.8 ppm more CO_2 per point (95% interval roughly ± 4 ppm). Grouped by score, the mean CO_2 is 484 ppm for low scores (5 or less, $n = 207$), 509 ppm for middle scores (6 to 8, $n = 287$) and 531 ppm for high scores (9 or more, $n = 170$). The link is therefore unlikely to be chance, but it is small: the proxy explains only about 3% of the variation in CO_2 ($R^2 = 0.027$). For comparison, work versus home alone explains 0.2% ($R^2 = 0.002$), and the two together 2.9%.

The direction of the link cannot be established from these data. Participants who feel tired may have been in busier rooms, but stuffy air can also make people feel more tired.

5 Limitations

A few things limit what this study can show.

The study is small and short. Twenty-two people filled in the surveys, 20 of them have sensor data, and only 10 have sensors at both home and work. Everything was recorded over 17 days in summer, so nothing can be said about winter. All participants work in sport facilities, so the results may not apply to other people.

The sensors measure one spot in a room, not what a person breathes while moving around, and they were only checked with simple plausibility limits. Work or home was taken from the folder names of the sensor files. The comparison is made at survey times, and 49 of the 713 surveys could not be matched to a sensor. In the continuous data, which includes the nights, CO_2 is higher at home (566 against 448 ppm), so "higher at work" depends on when you look.

Occupancy and activity were not measured. The activity score is a guess made from tiredness and productivity, and tiredness could also be caused by bad air, so the link found is no proof of an effect. Finally, the 664 observations come from only 20 people, so they are not independent, many correlations were looked at without any correction, and CO_2 rarely got high (8 observations above 1000 ppm). The results describe this group and period and should not be taken as general or causal.

6 Discussion and Conclusion

The results fit the main expectation: at survey times, CO_2 is higher in sport facilities than at home, probably because of occupancy and physical activity. The average difference is small (about 12 ppm), and both environments stay mostly within acceptable ranges. Readings above 800 ppm are slightly more frequent in sport facilities (4.6% against 3.1%). That could matter to facility managers during busy sessions, but the gap is small and would need occupancy data to confirm. The continuous record includes the nights at home and shows higher CO_2 there (566 against 448 ppm), so the comparison covers the time participants were awake and answering surveys, not total daily exposure.

The second research question, whether activity intensity affects CO_2, can only be answered in part. The proxy built from tiredness and productivity is weakly related to CO_2 and explains about 3% of its variation. Because activity was approximated with a questionnaire score and the direction of the link is open, this is a hint and not a finding. Measuring activity directly, for example with movement sensors or session logs, would be

needed to settle it.

Particulate matter shows the opposite pattern, with PM2.5 slightly higher at home. This shows the two environments present different types of air quality risk. Sport facilities raise CO_2 through exertion and occupancy, while homes may contribute more to fine particle exposure through domestic sources.

The weak correlations between air quality and subjective well-being were somewhat unexpected but plausible. Most CO_2 readings were in the mild range where effects on perception are subtle. The link between perceived warmth and tiredness ($\rho \approx 0.30$) is the clearest subjective result. Perceived warmth follows measured temperature ($\rho \approx 0.51$), but measured temperature itself is only weakly related to tiredness ($\rho \approx 0.07$). The evidence that warmer conditions increase fatigue is therefore indirect. It points to thermal management as something worth looking at in sport facilities, but it does not show an effect.

Methodologically, the two key decisions, using Spearman correlation for ordinal survey data and using linear interpolation rather than window averaging for timestamp alignment, were important for producing statistically valid results. Future work could improve further by collecting occupancy counts directly, increasing sensor sampling frequency, and extending the study period to capture seasonal variation.

In summary, this project provided a data-driven picture of indoor air quality in sport facilities. CO_2 and PM2.5 vary in opposite directions between work and home, temporal patterns track occupancy hours, and subjective well-being shows only weak ties to short-term air quality fluctuations at these moderate exposure levels.